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Section: ICS-01-402A

SECURITY: FACE RECOGNITION

Throughout the project, I handled several important tasks related to the backend and database integration of the system. I worked on adding edits with my groupmates to the JSON file code where we included the student information needed for the system output. I and Kazandra also integrated the JSON file into the FastAPI framework and helped in coding the `api.py` file in the backend. Aside from that, I worked on the SQLite3 database setup including the `seed.py`, `students.db`, and other related backend files to ensure the system could properly store and retrieve student data.

During the development process, I encountered different challenges that tested my problem-solving skills. One issue happened when we accidentally used the Python 13 environment instead of Python 10, which caused several compatibility errors in the project. Another challenge was the slow loading of Streamlit, especially during testing and debugging. At first, the facial recognition feature also failed to detect my face properly, but after troubleshooting and adjusting the setup, it finally worked successfully. One of the major issues I solved with the help of my groupmate Kazandra, was the duplicate student ID error in the database. Because of a mistake in the JSON file, duplicate student IDs were entered even though the SQLite database only accepts unique IDs as primary keys. This caused my classmate's face data not to be detected properly. I fixed the issue by correcting the student IDs and Kazandra helped me in updating the database, which made the system function correctly again.

Through this project, I discovered and learned many new things about system development and facial recognition technology. I learned how facial recognition libraries work and understood the logic behind how the system detects and identifies faces. I also gained knowledge about integrating FastAPI into a system and how backend APIs communicate with the frontend. Additionally, I learned that SQLite databases only allow unique primary keys, which helped me better understand proper database management and data validation.

My personal contribution to the project focused mainly on improving the backend system. I helped fix the duplicate student ID issue, solved small bugs and coding errors, and helped manage important backend files such as `database.py`, `seed.py`, and `students.db`. These contributions helped improve the overall functionality and stability of the system.

Overall, my experience working with the team was good and productive. I learned many new skills and techniques from my groupmates, especially in backend development and system integration and it made the project more manageable because everyone contributed their ideas and efforts to complete the system successfully.